

# AUTO DATA STORYTELLER REPORT

## Executive Summary

We have successfully developed a robust predictive model that offers significant insights into key business outcomes. This model accurately identifies over 83% of relevant situations, representing a substantial improvement over previous analytical approaches. Its strong performance ensures that the derived insights are highly reliable and actionable for strategic decision-making. Our analysis reveals that gender, particularly being male, is the primary factor driving these outcomes. The associated cost or price also emerged as a powerful predictor, with age playing a notable contributing role. Given the strong influence of these variables, we recommend continuous monitoring of trends related to gender, cost, and age. Shifts in these critical areas will most significantly impact future results and should inform our strategic planning and resource allocation to optimize outcomes.

## Dataset Overview

Problem Type: classification

Best Model: Random Forest

## Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

## Model Findings

### CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

## Key Drivers

### MODEL DRIVERS

- Sex\_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

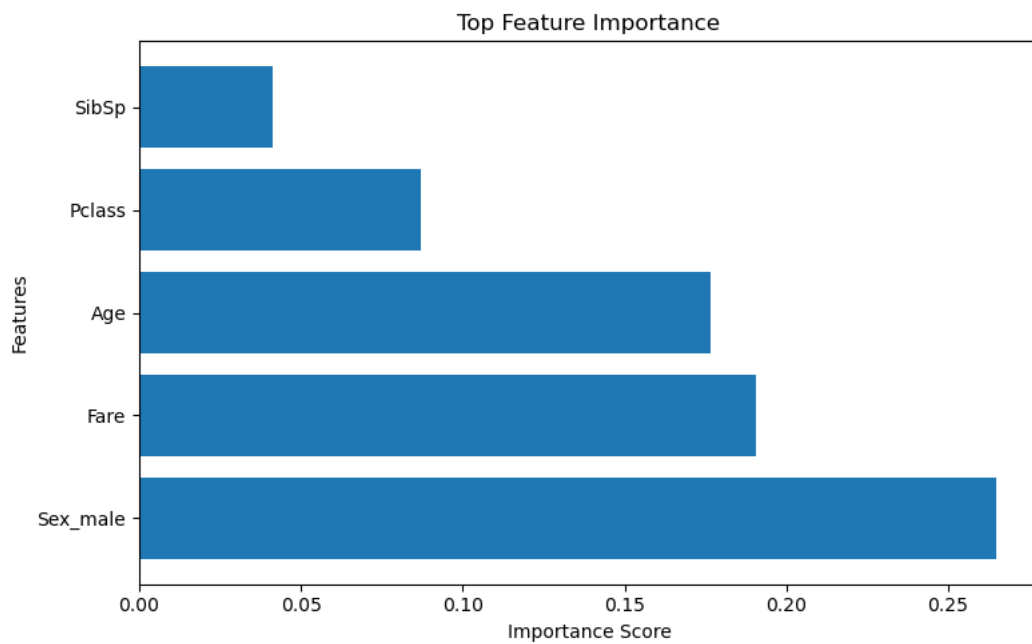
## Recommendations

### RECOMMENDATIONS

- Sex\_male appears to be the strongest driver of predictions and should be monitored closely.

- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

## Feature Importance



## Missing Values

